

Drayan Mathematical Series

Geometry #1

Circles and Triangles, Quadrilaterals,
Loci, Concurrence, and Olympiad Problems

This handout is a curated compilation of geometry problems from the *Indian Regional Mathematical Olympiad* (RMO) and its successor, the *Central RMO* (CRMO), spanning the years 1990 to 2019. Problems cover classical Euclidean geometry: triangle centres, circle geometry, cyclic quadrilaterals, angle chasing, locus arguments, and concurrence results. Problems are organised chronologically by year. No solutions are provided; the reader is encouraged to approach each problem independently before consulting external resources.

Drayan Mathematical Series · Geometry #1 · All rights reserved

Contents

1	Triangle Geometry	2
1.1	Triangle Centres and Basic Properties	2
1.2	Angle Bisectors and the Incircle	3
1.3	Orthocentre, Circumcentre, and Special Points	4
2	Circle Geometry	5
2.1	Cyclic Quadrilaterals and Power of a Point	5
2.2	Tangent Lines, Radical Axes, and Inversion	6
2.3	Reflections and Isosceles/Equilateral Configurations	7
3	Quadrilaterals and Polygons	8
3.1	Special Quadrilaterals	8
3.2	Midpoint and Ratio Problems	10
4	Locus, Construction, and Miscellaneous	10
4.1	Locus and Construction Problems	11
4.2	Hexagons and Area Relations	11
5	Comprehensive Olympiad Problems	11

§1 Triangle Geometry

1.1 Triangle Centres and Basic Properties

The problems in this subsection develop fluency with the classical triangle centres — circumcentre, incentre, centroid, and orthocentre — together with the fundamental metric relations that connect them.

Problem 1990 RMO P8

If the circumcentre and centroid of a triangle coincide, prove that it must be equilateral.

Problem 1990 RMO P5

Let P be any point inside a triangle ABC . The perimeter of the triangle is $AB+BC+CA = 2s$. Prove that

$$s < AP + BP + CP < 2s.$$

Problem 1991 RMO P1

Let P be an interior point of a triangle ABC and let AP, BP, CP meet the sides BC, CA, AB in D, E, F respectively. Show that

$$\frac{AP}{PD} = \frac{AF}{FB} + \frac{AE}{EC}.$$

Problem 1991 RMO P5

Take any point P_1 on the side BC of a triangle ABC and draw the following chain of lines:

$$P_1P_2 \parallel AC, \quad P_2P_3 \parallel BC, \quad P_3P_4 \parallel AB, \quad P_4P_5 \parallel CA, \quad P_5P_6 \parallel BC,$$

where P_2, P_5 lie on AB ; P_3, P_6 lie on CA ; and P_4 lies on BC . Show that P_6P_1 is parallel to AB .

Problem 1993 RMO P1

Let ABC be an acute-angled triangle and CD be the altitude through C . If $AB = 8$ and $CD = 6$, find the distance between the midpoints of AD and BC .

Problem 1997 RMO P1

Let P be an interior point of a triangle ABC and let BP and CP meet AC and AB in E and F respectively. If $[BPF] = 4$, $[BPC] = 8$, and $[CPE] = 13$, find $[AFPE]$.

Problem 2001 RMO P1

Let BE and CF be the altitudes of an acute triangle ABC with E on AC and F on AB . Let O be the point of intersection of BE and CF . Take any line KL through O with K on AB and L on AC . Suppose M and N are located on BE and CF respectively such that $KM \perp BE$ and $LN \perp CF$. Prove that $FM \parallel EN$.

Problem 2001 RMO P5

In a triangle ABC , D is a point on BC such that AD is the internal bisector of $\angle A$. Suppose $\angle B = 2\angle C$ and $CD = AB$. Prove that $\angle A = 72^\circ$.

Problem 2002 RMO P1

In an acute triangle ABC , points D, E, F are located on the sides BC, CA, AB such that

$$\frac{CD}{CE} = \frac{CA}{CB}, \quad \frac{AE}{AF} = \frac{AB}{AC}, \quad \frac{BF}{FD} = \frac{BC}{BA}.$$

Prove that AD, BE, CF are the altitudes of triangle ABC .

Problem 2005 RMO P5

In a triangle ABC , D is the midpoint of BC . If $\angle ADB = 45^\circ$ and $\angle ACD = 30^\circ$, determine $\angle BAD$.

Problem 2007 RMO P1

Let ABC be an acute-angled triangle; AD be the bisector of $\angle BAC$ with D on BC ; and BE be the altitude from B on AC . Show that $\angle CED > 45^\circ$.

Problem 2008 RMO P1

Let ABC be an acute-angled triangle; let D, F be the midpoints of BC, AB respectively. Let the perpendicular from F to AC and the perpendicular from B to BC meet in N . Prove that ND is the circumradius of $\triangle ABC$.

1.2 Angle Bisectors and the Incircle

Problem 1994 RMO P2

In a triangle ABC , the incircle touches the sides BC, CA, AB at D, E, F respectively. If the inradius is 4 units and BD, CE, AF are consecutive integers, find the sides of triangle ABC .

Problem 1995 RMO P1

In triangle ABC , K and L are points on side BC (K closer to B than L) such that $BC \cdot KL = BK \cdot CL$ and AL bisects $\angle KAC$. Show that $AL \perp AB$.

Problem 1996 RMO P1

The sides of a triangle are three consecutive integers and its inradius is 4. Find the circumradius.

Problem 1999 RMO P1

Prove that the inradius of a right-angled triangle with integer sides is an integer.

Problem 2009 RMO P1

Let ABC be a triangle in which $AB = AC$ and let I be its incentre. Suppose $BC = AB + AI$. Find $\angle BAC$.

Problem 2011 RMO P5

Let ABC be a triangle and let BB_1, CC_1 be respectively the bisectors of $\angle B, \angle C$ with B_1 on AC and C_1 on AB . Let E, F be the feet of perpendiculars drawn from A onto BB_1, CC_1 respectively. Suppose D is the point at which the incircle of ABC touches AB . Prove that $AD = EF$.

Problem 2012 RMO Region 1 P5

Let ABC be a triangle. Let BE and CF be internal angle bisectors of $\angle B$ and $\angle C$ with E on AC and F on AB . Suppose X is a point on segment CF such that $AX \perp CF$, and Y is a point on segment BE such that $AY \perp BE$. Prove that

$$XY = \frac{b + c - a}{2},$$

where $BC = a, CA = b, AB = c$.

1.3 Orthocentre, Circumcentre, and Special Points

Problem 2003 RMO P5

Suppose P is an interior point of a triangle ABC such that the ratios

$$\frac{d(A, BC)}{d(P, BC)}, \quad \frac{d(B, CA)}{d(P, CA)}, \quad \frac{d(C, AB)}{d(P, AB)}$$

are all equal, where $d(X, YZ)$ denotes the perpendicular distance from X to line YZ . Find the common value of these ratios.

Problem 2010 RMO P5

Let ABC be a triangle in which $\angle A = 60^\circ$. Let BE and CF be the bisectors of $\angle B$ and $\angle C$ with E on AC and F on AB . Let M be the reflection of A in line EF . Prove that M lies on BC .

Problem 2011 RMO P1

Let ABC be a triangle. Let D, E, F be points on segments BC, CA, AB respectively such that AD, BE, CF concur at the point K . Suppose

$$\frac{BD}{DC} = \frac{BF}{FA} \quad \text{and} \quad \angle ADB = \angle AFC.$$

Prove that $\angle ABE = \angle CAD$.

Problem 2011 RMO Advanced P1

Let ABC be an acute-angled scalene triangle with circumcentre O and orthocentre H . If M is the midpoint of BC , show that AO and HM intersect on the circumcircle of $\triangle ABC$.

Problem 2014 RMO Region 1 P5

Let ABC be an acute-angled triangle with orthocentre H . For any point P on the circumcircle of $\triangle ABC$, let Q be the point of intersection of line BH with line AP . Show that there is a unique point X on the circumcircle of $\triangle ABC$ such that for every P other than B, C , the circumcircle of $\triangle HPQ$ passes through X .

§2 Circle Geometry

2.1 Cyclic Quadrilaterals and Power of a Point

Problem 1992 RMO P4

$ABCD$ is a cyclic quadrilateral with $AC \perp BD$; AC meets BD at E . Prove that

$$EA^2 + EB^2 + EC^2 + ED^2 = 4R^2,$$

where R is the radius of the circumscribed circle.

Problem 1992 RMO P8

The cyclic octagon $ABCDEFGH$ has sides a, a, a, a, b, b, b, b respectively. Find the radius of the circle that circumscribes $ABCDEFGH$.

Problem 1994 RMO P6

Let AC and BD be two chords of a circle with centre O such that they intersect at right angles inside the circle at point M . Suppose K and L are midpoints of chords AB and CD respectively. Prove that $OKML$ is a parallelogram.

Problem 1998 RMO P1

Let $ABCD$ be a convex quadrilateral in which $\angle BAC = 50^\circ$, $\angle CAD = 60^\circ$, and $\angle BDC = 25^\circ$. If E is the point of intersection of AC and BD , find $\angle AEB$.

Problem 2000 RMO P5

The internal bisector of angle A in a triangle ABC with $AC > AB$ meets the circumcircle Γ in D . Join D to the centre O of Γ and suppose that DO meets AC in E (possibly when extended). Given that $BE \perp AD$, show that $AO \parallel BD$.

Problem 2002 RMO P5

The circumference of a circle is divided into eight arcs by a convex quadrilateral $ABCD$, with four arcs lying inside the quadrilateral and four outside. The lengths of the arcs lying inside are p, q, r, s in counter-clockwise order. Suppose $p + r = q + s$. Prove that $ABCD$ is a cyclic

quadrilateral.

Problem 2004 RMO P1

Consider a circle Γ with centre O and a line ℓ not intersecting Γ . Prove that there is a unique point Q on the perpendicular from O to ℓ such that for any point P on ℓ , the length PQ equals the length of the tangent from P to Γ .

Problem 2013 RMO Region 4 P1

Let ω be a circle with centre O . Let γ be another circle passing through O and intersecting ω at points A and B . A diameter CD of ω intersects γ at a point P different from O . Prove that $\angle APC = \angle BPD$.

Problem 2013 RMO Region 1 P1

Let ABC be an acute-angled triangle. The circle Γ with BC as diameter intersects AB and AC again at P and Q respectively. Determine $\angle BAC$ given that the orthocentre of $\triangle APQ$ lies on Γ .

2.2 Tangent Lines, Radical Axes, and Inversion

Problem 2013 RMO Region 3 P3

In an acute-angled triangle ABC with $AB < AC$, the circle ω touches AB at B and passes through C , intersecting AC again at D . Prove that the orthocentre of $\triangle ABD$ lies on ω if and only if it lies on the perpendicular bisector of BC .

Problem 2013 RMO Region 2 P2

In a triangle ABC , AD is the altitude from A and H is the orthocentre. Let K be the centre of the circle passing through D and tangent to BH at H . Prove that line DK bisects AC .

Problem 2015 RMO Region 4 P5

Two circles Γ and Σ intersect at two distinct points A and B . A line through B intersects Γ and Σ again at C and D respectively. Suppose that $CA = CD$. Show that the centre of Σ lies on Γ .

Problem 2015 RMO Region 2 P5

Two circles X and Y in the plane intersect at two distinct points A and B such that the centre of Y lies on X . Let C and D be on X and Y respectively, so that C, B, D are collinear. Let E on Y be such that $DE \parallel AC$. Show that $AE = AB$.

Problem 2015 RMO Region 3 P1

Two circles Γ and Σ , with centres O and P respectively, are such that P lies on Γ . Let A be a point on Σ and let M be the midpoint of AP . Let B be another point on Σ such that

$AB \parallel OM$. Prove that the midpoint of AB lies on Γ .

Problem 2016 RMO Delhi P1

Two circles ω_1, ω_2 intersect at points X, Y . Let P be an arbitrary point on ω_1 . Suppose the lines PX, PY meet ω_2 again at points A, B respectively. Prove that the circumcircles of all triangles PAB have the same radius.

Problem 2016 RMO Delhi P3

Two circles C_1 and C_2 intersect each other at points A and B . Their external common tangent (closer to B) touches C_1 at P and C_2 at Q . Let C be the reflection of B in line PQ . Prove that $\angle CAP = \angle BAQ$.

Problem 2017 RMO Odisha P5

Let Ω be a circle with a chord AB which is not a diameter. Let Γ_1 be a circle on one side of AB , tangent to AB at C and internally tangent to Ω at D . Let Γ_2 be a circle on the other side of AB , tangent to AB at E and internally tangent to Ω at F . Suppose line DC intersects Ω at $X \neq D$ and line FE intersects Ω at $Y \neq F$. Prove that XY is a diameter of Ω .

Problem 2018 RMO P1

Let ABC be a triangle with integer sides in which $AB < AC$. Let the tangent to the circumcircle of $\triangle ABC$ at A intersect line BC at D . Suppose AD is also an integer. Prove that $\gcd(AB, AC) > 1$.

Problem 2018 RMO Kerala P1

Let ABC be an acute-angled triangle and let D be an interior point of segment BC . Let the circumcircle of $\triangle ACD$ intersect AB at E (between A and B) and let the circumcircle of $\triangle ABD$ intersect AC at F (between A and C). Let O be the circumcentre of $\triangle AEF$. Prove that OD bisects $\angle EDF$.

2.3 Reflections and Isosceles/Equilateral Configurations

Problem 1998 RMO P4

Let ABC be a triangle with $AB = AC$ and $\angle BAC = 30^\circ$. Let A' be the reflection of A in line BC ; B' the reflection of B in line CA ; C' the reflection of C in line AB . Show that $A'B'C'$ is an equilateral triangle.

Problem 2000 RMO P1

Let AC be a line segment in the plane and B a point between A and C . Construct isosceles triangles PAB and QBC on one side of AC such that $\angle APB = \angle BQC = 120^\circ$, and an isosceles triangle RAC on the other side of AC such that $\angle ARC = 120^\circ$. Show that PQR is an equilateral triangle.

Problem 2013 RMO Region 3 P5

Let ABC be a triangle which is not right-angled. Define a sequence of triangles $A_iB_iC_i$ ($i \geq 0$) as follows: $A_0B_0C_0$ is the triangle ABC , and for $i \geq 0$, A_{i+1} , B_{i+1} , C_{i+1} are the reflections of the orthocentre of $\triangle A_iB_iC_i$ in the sides B_iC_i , C_iA_i , A_iB_i respectively. Assume that $\angle A_m = \angle A_n$ for some distinct natural numbers m, n . Prove that $\angle A = 60^\circ$.

Problem 2013 RMO Region 4 P5

In a triangle ABC , let H denote its orthocentre. Let P be the reflection of A with respect to BC . The circumcircle of $\triangle ABP$ intersects line BH again at Q , and the circumcircle of $\triangle ACP$ intersects line CH again at R . Prove that H is the incentre of $\triangle PQR$.

§3 Quadrilaterals and Polygons

3.1 Special Quadrilaterals

Problem 1990 RMO P3

A square sheet of paper $ABCD$ is folded so that B falls on the midpoint M of CD . Prove that the crease will divide BC in the ratio $5 : 3$.

Problem 1992 RMO P5

$ABCD$ is a quadrilateral and P, Q are the midpoints of CD, AB respectively. AP and DQ meet at X , and BP and CQ meet at Y . Prove that $[ADX] + [BCY] = [PXQY]$.

Problem 1993 RMO P4

Let $ABCD$ be a rectangle with $AB = a$ and $BC = b$. Suppose r_1 is the radius of the circle passing through A and B and touching CD , and r_2 is the radius of the circle passing through B and C and touching AD . Show that

$$r_1 + r_2 \geq \frac{5}{8}(a + b).$$

Problem 1997 RMO P4

In a quadrilateral $ABCD$, $AB \parallel CD$ and the diagonals AC and BD are perpendicular to each other. Show that

- (a) $AD \cdot BC \geq AB \cdot CD$,
- (b) $AD + BC \geq AB + CD$.

Problem 1999 RMO P3

Let $ABCD$ be a square and M, N points on sides AB, BC respectively such that $\angle MDN = 45^\circ$. If R is the midpoint of MN , show that $RP = RQ$ where P, Q are the points of intersection of AC with the lines MD, ND .

Problem 2004 RMO P5

Let $ABCD$ be a quadrilateral, X and Y the midpoints of AC and BD respectively, and lines through X and Y parallel to BD and AC meet at O . Let P, Q, R, S be the midpoints of AB, BC, CD, DA respectively. Prove that

- (a) $[APOS]$ and $[APXS]$ have the same area,
- (b) $[APOS], [BQOP], [CROQ], [DSOR]$ have the same area.

Problem 2005 RMO P1

Let $ABCD$ be a convex quadrilateral. P, Q, R, S are the midpoints of AB, BC, CD, DA respectively such that triangles AQR and CSP are equilateral. Prove that $ABCD$ is a rhombus and find its angles.

Problem 2006 RMO P5

Let $ABCD$ be a quadrilateral in which $AB \parallel CD$, $AB \perp AD$, and $AB = 3CD$. If the area of the quadrilateral is 4 and a circle can be drawn touching all four sides, find the radius of that circle.

Problem 2007 RMO P5

A trapezium $ABCD$, in which $AB \parallel CD$, is inscribed in a circle with centre O . Suppose the diagonals AC and BD intersect at M , and $OM = 2$.

- (a) If $\angle AMB = 60^\circ$, find, with proof, the difference between the lengths of the parallel sides.
- (b) If $\angle AMD = 60^\circ$, find, with proof, the difference between the lengths of the parallel sides.

Problem 2011 RMO Advanced P5

Let $ABCD$ be a convex quadrilateral. Let E, F, G, H be the midpoints of AB, BC, CD, DA respectively. If AC, BD, EG, FH all concur at a point O , prove that $ABCD$ is a parallelogram.

Problem 2012 RMO Region 6 P1

Let $ABCD$ be a convex quadrilateral such that $\angle ADC = \angle BCD > 90^\circ$. Let E be the intersection of AC and the line through B parallel to AD ; let F be the intersection of BD and the line through A parallel to BC . Prove that $EF \parallel CD$.

Problem 2014 RMO Region 4 P1

Let $ABCD$ be an isosceles trapezium having an incircle, with $AB \parallel CD$. Let CE be the perpendicular from C on AB . Prove that $CE^2 = AB \cdot CD$.

Problem 2015 RMO Mumbai P1

Let $ABCD$ be a convex quadrilateral with $AB = a$, $BC = b$, $CD = c$, $DA = d$. Suppose

$$a^2 + b^2 + c^2 + d^2 = ab + bc + cd + da,$$

and the area of $ABCD$ is 60 sq. units. If one diagonal has length 30 units, determine the length of the other diagonal.

3.2 Midpoint and Ratio Problems**Problem 2003 RMO P1**

Let ABC be a triangle with $AB = AC$ and $\angle CAB = 90^\circ$. Suppose M and N are points on the hypotenuse BC such that $BM^2 + CN^2 = MN^2$. Prove that $\angle MAN = 45^\circ$.

Problem 2012 RMO Region 1 P1

Let ABC be a triangle and D a point on segment BC such that $DC = 2BD$. Let E be the midpoint of AC . Let AD and BE intersect at P . Determine the ratios $BP : PE$ and $AP : PD$.

Problem 2012 RMO Region 2 P5

Let ABC be a triangle. Let D, E be points on segment BC such that $BD = DE = EC$. Let F be the midpoint of AC . Let BF intersect AD at P and AE at Q . Determine $BP : PQ$.

Problem 2012 RMO Region 3 P5

Let ABC be a triangle. Let D, E be points on segment BC such that $BD = DE = EC$. Let F be the midpoint of AC . Let BF intersect AD at P and AE at Q . Determine the ratio of the area of $\triangle APQ$ to that of quadrilateral $PDEQ$.

Problem 2012 RMO Region 4 P5

Let ABC be a triangle. Let E be a point on segment BC such that $BE = 2EC$. Let F be the midpoint of AC . Let BF intersect AE at Q . Determine $BQ : QF$.

Problem 2013 RMO Mumbai P4

In a triangle ABC , points D and E are on segments BC and AC respectively such that $BD = 3DC$ and $AE = 4EC$. Point P is on line ED such that D is the midpoint of EP . Lines AP and BC intersect at S . Find the ratio $BS : SD$.

§4 Locus, Construction, and Miscellaneous

4.1 Locus and Construction Problems

Problem 2006 RMO P1

Let ABC be an acute-angled triangle and let D, E, F be the feet of the perpendiculars from A, B, C to BC, CA, AB respectively. Let the perpendiculars from F to CB, CA, AD, BE meet them in P, Q, M, N respectively. Prove that P, Q, M, N are collinear.

Problem 2009 RMO P5

A convex polygon is such that the distance between any two vertices does not exceed 1.

- (i) Prove that the distance between any two points on the boundary of the polygon does not exceed 1.
- (ii) If X and Y are two distinct points inside the polygon, prove that there exists a point Z on the boundary such that $XZ + YZ \leq 1$.

Problem 2016 RMO Maharashtra & Goa P4

Let $\triangle ABC$ be scalene with BC as the largest side. Let D be the foot of the perpendicular from A on BC . Let K, L be chosen on lines AB and AC respectively such that D is the midpoint of KL . Prove that B, K, C, L are concyclic if and only if $\angle BAC = 90^\circ$.

Problem 2017 RMO Odisha P1

Let $\angle AOB$ be a given angle less than 180° and let P be an interior point of the angular region determined by $\angle AOB$. Show, with proof, how to construct using only ruler and compass a line segment CD passing through P such that C lies on ray OA , D lies on ray OB , and $CP : PD = 1 : 2$.

Problem 2019 RMO Maharashtra & Goa P2

Given a circle τ , let P be a point in its interior and ℓ a line through P . Construct, with proof using ruler and compass, all circles which pass through P , are tangent to τ , and whose centre lies on line ℓ .

4.2 Hexagons and Area Relations

Problem 2010 RMO P1

Let $ABCDEF$ be a convex hexagon in which diagonals AD, BE, CF are concurrent at O . Suppose $[OAF]$ is the geometric mean of $[OAB]$ and $[OEF]$, and $[OBC]$ is the geometric mean of $[OAB]$ and $[OCD]$. Prove that $[OED]$ is the geometric mean of $[OCD]$ and $[OEF]$.

§5 Comprehensive Olympiad Problems

The following problems are drawn from selected years of the RMO and CRMO, synthesising all techniques from the preceding sections. The reader is expected to draw on a broad repertoire of methods.

Problem 2012 RMO Mumbai P4

H is the orthocentre of an acute-angled triangle ABC . A point E is taken on segment CH such that $\triangle ABE$ is right-angled. Prove that the area of $\triangle ABE$ is the geometric mean of the areas of $\triangle ABC$ and $\triangle ABH$.

Problem 2012 RMO Mumbai P7

On the extension of chord AB of a circle with centre O , a point X is taken and tangents XC and XD to the circle are drawn from it, with C and D on the circle. Let E be the midpoint of CD . If $\angle OEB = 140^\circ$, determine, with proof, the magnitude of $\angle AOB$.

Problem 2012 RMO Region 6 P5

Let AL and BK be the angle bisectors in a non-isosceles triangle ABC , where L lies on BC and K lies on AC . The perpendicular bisector of BK intersects line AL at M . Point N lies on line BK such that $LN \parallel MK$. Prove that $LN = NA$.

Problem 2013 RMO Region 1 P5

Let ABC be a triangle with $\angle A = 90^\circ$ and $AB = AC$. Let D and E be points on segment BC such that $BD : DE : EC = 3 : 5 : 4$. Prove that $\angle DAE = 45^\circ$.

Problem 2013 RMO Region 2 P4

Let ABC be a triangle with $\angle A = 90^\circ$ and $AB = AC$. Let D and E be points on segment BC such that $BD : DE : EC = 1 : 2 : \sqrt{3}$. Prove that $\angle DAE = 45^\circ$.

Problem 2013 RMO Mumbai P1

Let ABC be an isosceles triangle with $AB = AC$ and let Γ denote its circumcircle. A point D is on arc AB of Γ not containing C . A point E is on arc AC of Γ not containing B . If $AD = CE$, prove that $BE \parallel AD$.

Problem 2014 RMO Region 2 P1

In an acute-angled triangle ABC , $\angle ABC$ is the largest angle. The perpendicular bisectors of BC and BA intersect AC at X and Y respectively. Prove that the circumcentre of $\triangle ABC$ is the incentre of $\triangle BXY$.

Problem 2014 RMO Region 2 P5

Let ABC be a triangle with $AB > AC$. Let P be a point on line AB beyond A such that $AP + PC = AB$. Let M be the midpoint of BC and Q the point on side AB such that $CQ \perp AM$. Prove that $BQ = 2AP$.

Problem 2014 RMO Region 3 P1

Let ABC be a triangle with $\angle ABC$ as the largest angle. Let R be its circumcentre. Let the circumcircle of $\triangle ARB$ cut AC again at X . Prove that $RX \perp BC$.

Problem 2014 RMO Region 3 P5

Let ABC be a triangle and let X be a point on BC such that $AX = AB$. Let AX meet the circumcircle ω of $\triangle ABC$ again at D . Prove that the circumcentre of $\triangle BDY$ lies on ω .

Problem 2014 RMO Region 4 P5

Let ABC be a triangle and I its incentre. Let the incircle of $\triangle ABC$ touch BC at D . Let the incircle of $\triangle ABD$ touch AB at E , and let the incircle of $\triangle ACD$ touch AC at F . Prove that B, E, I, F are concyclic.

Problem 2014 RMO Mumbai P3

Let ABC be an acute-angled triangle in which $\angle ABC$ is the largest angle. Let O be its circumcentre. The perpendicular bisectors of BC and AB meet AC at X and Y respectively. The internal angle bisectors of $\angle AXB$ and $\angle BYC$ meet AB and BC at D and E respectively. Prove that $BO \perp AC$ if $DE \parallel AC$.

Problem 2014 RMO Mumbai P6

Let D, E, F be the points of contact of the incircle of an acute-angled triangle ABC with BC, CA, AB respectively. Let I_1, I_2, I_3 be the incentres of triangles AFE, BDF, CED respectively. Prove that lines I_1D, I_2E, I_3F are concurrent.

Problem 2015 RMO Region 1 P1

In a cyclic quadrilateral $ABCD$, let the diagonals AC and BD intersect at X . Let the circumcircles of $\triangle AXD$ and $\triangle BXC$ intersect again at Y . If X is the incentre of $\triangle ABY$, show that $\angle CAD = 90^\circ$.

Problem 2015 RMO Region 1 P5

Let ABC be a right triangle with $\angle B = 90^\circ$. Let E and F be the midpoints of AB and AC respectively. Suppose the incentre I of $\triangle ABC$ lies on the circumcircle of $\triangle AEF$. Find the ratio $BC : AB$.

Problem 2015 RMO Region 2 P1

Let ABC be a triangle. Let B' and C' denote the reflections of B and C in the internal angle bisector of $\angle A$. Show that triangles ABC and $AB'C'$ have the same incentre.

Problem 2015 RMO Region 3 P5

Let ABC be a triangle with circumcircle Γ and incentre I . Let the internal angle bisectors of $\angle A, \angle B, \angle C$ meet Γ in A', B', C' respectively. Let $B'C'$ intersect AA' at P and AC at Q . Let BB' intersect AC at R . Suppose quadrilateral $PIRQ$ is a kite: $IP = IR$ and $QP = QR$. Prove that $\triangle ABC$ is equilateral.

Problem 2015 RMO Region 4 P1

Let ABC be a triangle. Let B' denote the reflection of B in the internal angle bisector ℓ of $\angle A$. Show that the circumcentre of $\triangle CB'I$ lies on line ℓ , where I is the incentre of $\triangle ABC$.

Problem 2015 RMO Mumbai P5

Let ABC be a right-angled triangle with $\angle B = 90^\circ$ and let BD be the altitude from B to AC . Draw $DE \perp AB$ and $DF \perp BC$. Let P, Q, R, S be the incentres of triangles DFC, DBF, DEB , and DAE respectively. Suppose S, R, Q are collinear. Prove that P, Q, R, D lie on a circle.

Problem 2016 RMO Mumbai P1

Let ABC be a right-angled triangle with $\angle B = 90^\circ$. Let I be the incentre of $\triangle ABC$. Draw a line perpendicular to AI at I and let it intersect line CB at D . Prove that $CI \perp AD$ and that

$$ID = \sqrt{b(b-a)},$$

where $BC = a$ and $CA = b$.

Problem 2016 RMO Mumbai P5

Let ABC be a triangle with centroid G . Let the circumcircle of $\triangle AGB$ intersect line BC at X (different from B), and the circumcircle of $\triangle AGC$ intersect line BC at Y (different from C). Prove that G is the centroid of $\triangle AXY$.

Problem 2016 RMO Telangana P1

Let ABC be a right-angled triangle with $\angle B = 90^\circ$. Let I be the incentre of $\triangle ABC$. Suppose AI is extended to meet BC at F . The perpendicular to AI at I is extended to meet AC at E . Prove that $IE = IF$.

Problem 2016 RMO Telangana P5

Let ABC be a right-angled triangle with $\angle B = 90^\circ$. Let AD be the bisector of $\angle A$ with D on BC . Let the circumcircle of $\triangle ACD$ intersect AB again at E , and let the circumcircle of $\triangle ABD$ intersect AC again at F . Let K be the reflection of E in line BC . Prove that $FK = BC$.

Problem 2016 RMO Karnataka & WB P1

Let ABC be a triangle and D be the midpoint of BC . Suppose the angle bisector of $\angle ADC$ is tangent to the circumcircle of $\triangle ABD$ at D . Prove that $\angle A = 90^\circ$.

Problem 2016 RMO Karnataka & WB P5

Let ABC be a right-angled triangle with $\angle B = 90^\circ$. Let I be the incentre of $\triangle ABC$. Extend AI and CI to intersect BC at D and AB at E respectively. Draw a line perpendicular to AI at I to meet AC at J , and a line perpendicular to CI at I to meet AC at K . Suppose $DJ = EK$. Prove that $BA = BC$.

Problem 2016 RMO Odisha P1

Let ABC be an isosceles triangle with $AB = AC$. Let Γ be its circumcircle and O its centre. Let CO meet Γ in D . Draw a line parallel to AC through D and let it intersect AB at E . Suppose $AE : EB = 2 : 1$. Prove that $\triangle ABC$ is equilateral.

Problem 2016 RMO Odisha P5

Let ABC be a triangle, AD an altitude, and AE a median. Assume B, D, E, C lie in that order on line BC . Suppose the incentre of $\triangle ABE$ lies on AD and the incentre of $\triangle ADC$ lies on AE . Find, with proof, the angles of $\triangle ABC$.

Problem 2018 RMO P6

Let ABC be an acute-angled triangle with $AB < AC$. Let I be the incentre and D, E, F the points where the incircle touches BC, CA, AB respectively. Let BI and CI meet line EF at Y and X respectively. Assume both X and Y are outside $\triangle ABC$. Prove:

- (i) B, C, Y, X are concyclic.
- (ii) I is also the incentre of $\triangle DYX$.

Problem 2018 RMO Kerala P5

In a cyclic quadrilateral $ABCD$ with circumcentre O , the diagonals AC and BD intersect at X . Let the circumcircles of $\triangle AXD$ and $\triangle BXC$ intersect at Y . Let the circumcircles of $\triangle AXB$ and $\triangle CXD$ intersect at Z . If O lies inside $ABCD$ and the points O, X, Y, Z are all distinct, prove that O, X, Y, Z lie on a circle.

Problem 2019 RMO P2

Let ABC be a triangle with circumcircle Ω and centroid G . Extend AG, BG, CG to meet Ω again at A_1, B_1, C_1 respectively. Suppose

$$\angle BAC = \angle A_1B_1C_1, \quad \angle ABC = \angle A_1C_1B_1, \quad \angle ACB = \angle B_1A_1C_1.$$

Prove that $\triangle ABC$ and $\triangle A_1B_1C_1$ are equilateral.

Problem 2019 RMO P5

In an acute-angled triangle ABC , let H be the orthocentre and D, E, F the feet of the altitudes from A, B, C respectively. Let L, M, N be the midpoints of AH, EF, BC respectively. Let X, Y be the feet of altitudes from L, N onto line DF respectively. Prove that $XM \perp MY$.

Problem 2019 RMO Maharashtra & Goa P6

Let k be a positive real number. In the XY -plane let S be the set of all points of the form $(x, x^2 + k)$ where $x \in \mathbb{R}$. Let C be the set of all circles whose centre lies in S and which are tangent to the x -axis. Find the minimum value of k such that any two circles in C have at least one point of intersection.

End of Drayan Geometry #1

Drayan Mathematical Series · For instructional use only. No solutions are included in this document.